

Shubhankar Tiwari

Software Engineer / Backend Systems & Artificial Intelligence

+91 9910608519
st3907@columbia.edu
shubhankar-tiwari.vercel.app
shubhankar-tiwari-514040165/
shubhankartiwari99
tiwarishubhankar@gmail.com

Professional Summary

Software Engineer with **4+ years of experience** building enterprise backend systems and production software using **Java, Spring Boot, REST APIs, SQL, OpenShift, and CI/CD**. Incoming M.S. in Artificial Intelligence student at Columbia University with hands-on experience developing AI systems for LLM evaluation, model governance, and reliability. Passionate about combining rigorous software engineering with a deeper theoretical understanding of modern AI to build next-generation intelligent systems.

Experience

- Apr 2022 – Jul 2026
- Software Engineer, Bank of America, Hyderabad, India
- Built and maintained enterprise backend services for the CashPro platform using **Java, Spring Boot, REST APIs, SQL, and OpenShift**.
 - Owned application deployments on OpenShift, CI/CD pipeline configuration, release validation, and production stabilization across **17+ release cycles**.
 - Served as the **Subject Matter Expert (SME) for CashPro Search**, providing domain expertise, technical guidance, production troubleshooting, and knowledge transfer across engineering, QA, and release teams.
 - Contributed to modernization initiatives including **JDK 17, Spring 6**, containerization, cloud enablement, and SonarQube-driven quality improvements across **13 CashPro services**.
 - Diagnosed and resolved **50+ production issues** involving SSL certificates, database connectivity, Drools dependencies, deployment failures, privacy-related defects, and application functionality issues.
- Aug 2021 – Mar 2022
- Product Development Trainee, HighRadius Technologies
- Built backend modules for **Order-to-Cash and Treasury Systems** using **Java, JavaScript, and Python**, contributing to Agile sprint delivery and enterprise API integrations.

Projects

- 2025 – Present
- AI Reliability Platform for LLM Evaluation, Python, FastAPI, Information Theory
- Built an **evaluation and reliability platform** for large language models using entropy-aware routing, adaptive regeneration policies, and behavioral evaluation workflows.
 - Engineered an adversarial noise injector and asynchronous orchestration pipeline to evaluate multi-model stability across **Llama, Mistral, and Qwen**.
 - Conducted a controlled cultural-alignment study using **65 live inferences** across a 3-tier prompt gradient to analyze localization behavior and entropy collapse patterns.
 - Implemented a **9-category guardrail classifier** with severity-based escalation logic for self-harm, jailbreak, and extremism detection.
 - Designed **55+ automated regression tests**, deployment registries, and behavioral snapshot validation workflows for release gating.
- 2025 – Present
- Drift-Aware Fraud Detection & Model Governance, Python, XGBoost, FastAPI
- Built a **drift-aware ML governance pipeline** using PSI drift detection, threshold-aware promotion policies, and business-loss-based evaluation.
 - Implemented behavioral regression detection and CI-gated model promotion workflows to prevent misleading candidate models from deployment.

Education

- Aug 2026 – Dec 2027 (Expected)
- M.S. in Artificial Intelligence, Columbia Engineering, New York, NY
- 2018 – 2022
- B.Tech, Computer Science and Engineering, SRM Institute of Science and Technology, 94.4%

Recognition

- Awards
- Global Recognition Awards (Gold, Silver, Bronze) at Bank of America for backend delivery, cloud enablement, OpenShift migration, production stabilization, and release ownership across the CashPro platform.
- Kaggle
- Notebooks Expert — Personal Best #707; 10 Bronze Medals; Machine Learning, Deep Learning, and Computer Vision certified.

Technical Skills

- Backend
- Java, Spring Boot, REST APIs, SQL, OpenShift, Jenkins, Ansible, Linux
- AI/ML
- Python, PyTorch, Transformers, Scikit-learn, XGBoost
- ML Systems
- LLM Evaluation, AI Reliability, Model Validation, Behavioral Testing, Experimentation Frameworks, Model Governance